PROBLEM 1.3

Half-Life • Nuclear Decay • Space RTG

Plutonium-239 Radioactive Decay & Decay Constant

Original statement · ProblemsCh1.pdf

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Plutonium 239, virtually nonexistent in nature, is one of the radioactive materials used in the production of nuclear weapons, and is a byproduct of the generation of power in a nuclear reactor. Its half-life (i.e., the time it takes for half of the radioactive isotope to decay) is approximately 24,000 years.

Using $\frac{dN}{dt} = -kN$ as an appropriate model for radioactive decay, where $N = N(t)$ is the number of isotopes at time t, what is the value of k that should be used for this isotope?

PHASE 2: FUNDAMENTAL FORMULATION

Radioactive disintegration follows the first-order linear ODE $\frac{dN}{dt} = -kN$ ($k > 0$). By definition of half-life $t_{1/2} = 24{,}000$ yr, the isotope quantity at $t = t_{1/2}$ satisfies $N(t_{1/2}) = \frac{1}{2}N_0$.

PHASE 3: STEP-BY-STEP DERIVATION

Separate variables and integrate from $t = 0$ ($N = N_0$) to t:

$$\int_{N_0}^{N(t)} \frac{d\xi}{\xi} = \int_0^t -k\, ds \implies \ln\left(\frac{N(t)}{N_0}\right) = -kt \implies N(t) = N_0 e^{-kt}$$

Substitute $t = t_{1/2}$ and $N(t_{1/2}) = \frac{1}{2}N_0$:

$$N_0 e^{-kt_{1/2}} = \frac{1}{2}N_0 \implies e^{-kt_{1/2}} = 2^{-1}$$

Apply the natural logarithm to both sides:

$$\ln\left(e^{-kt_{1/2}}\right) = \ln\left(2^{-1}\right) \implies -kt_{1/2} = -\ln 2 \implies \mathbf{k = \frac{\ln 2}{t_{1/2}}}$$

Numerical evaluation with $t_{1/2} = 24{,}000$ years and $\ln 2 \approx 0.69314718$:

$$k = \frac{0.69314718}{24{,}000}\ \text{yr}^{-1} \approx \mathbf{2.8881 \times 10^{-5}\ \text{yr}^{-1}}$$

In SI units ($1\ \text{yr} \approx 365.25 \times 86400\ \text{s} = 3.15576 \times 10^{7}\ \text{s}$):

$$k = \frac{2.8881 \times 10^{-5}}{3.15576 \times 10^{7}}\ \text{s}^{-1} \approx \mathbf{9.1519 \times 10^{-13}\ \text{s}^{-1}}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$k = \frac{\ln 2}{24{,}000} \approx 2.8881 \times 10^{-5}\ \text{yr}^{-1} \approx 9.1519 \times 10^{-13}\ \text{s}^{-1}$$

Physical Interpretation: The extremely minute decay constant means that Plutonium-239 remains radioactive, essentially at constant activity, over many centuries (after $10\,t_{1/2} = 240{,}000$ years its activity is still $(1/2)^{10} \approx 0.098\%$ of the initial value). Note that space-probe RTGs (Voyager, Cassini, Curiosity) use the shorter-lived ^{238}Pu ($t_{1/2} \approx 87.7$ years), not ^{239}Pu.